Genomics and Databases – IMBM5

# CEDNIK Syndrome

Data Analysis for Corpasome family


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# CEDNIK Syndrome

Neurocutaneous syndrome, characterized by **ce**rebral **d**ysgenesis, **n**europathy, **i**chthyosis, and **k**eratoderma, or **CEDNIK** syndrome (609528) for short, is an autosomal recessive genetic disorder.

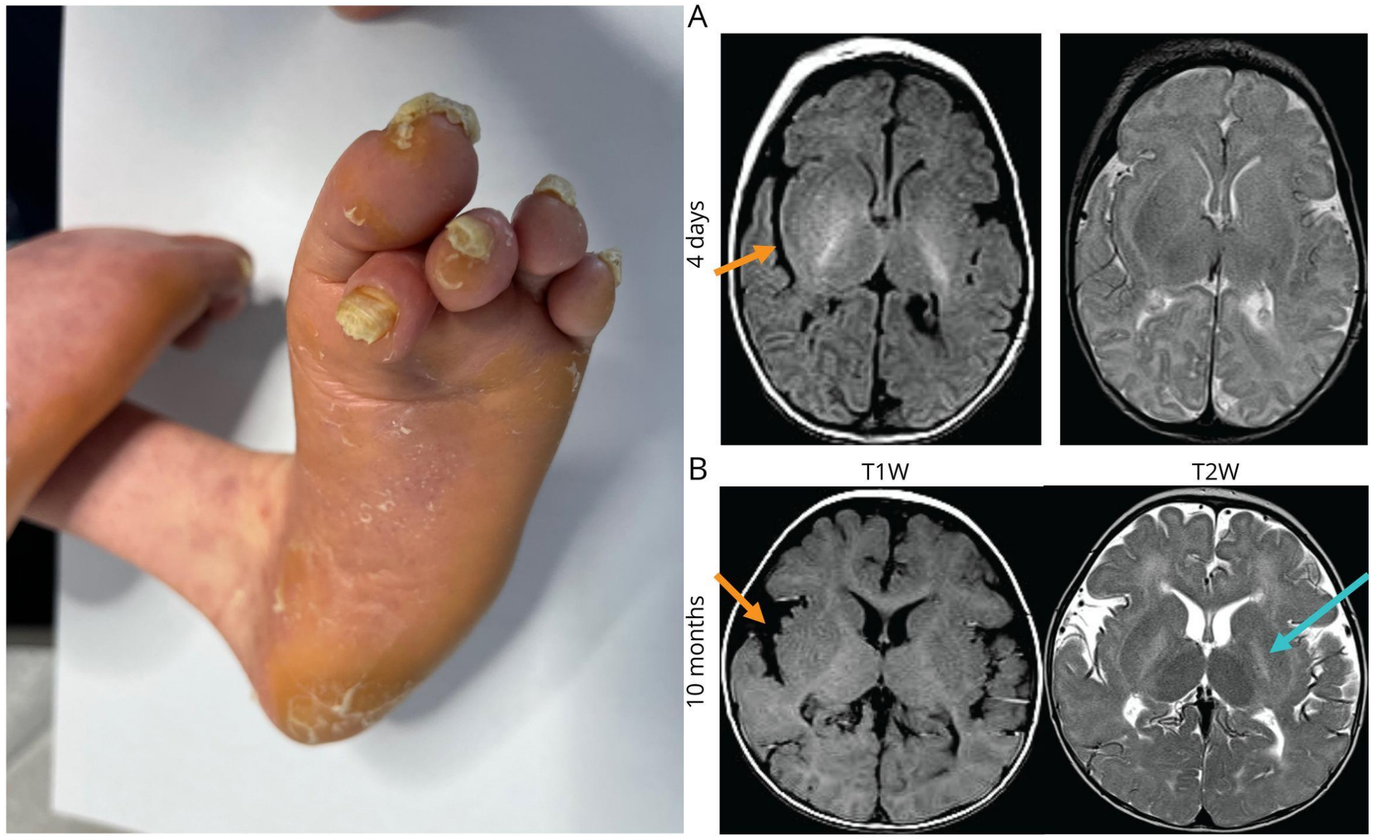



**Figure 1:** Skin Keratoderma and ichthyosis
**Figure 2:** Brain MRI of Patient 1 Shows Corpus Callosum Dysgenesis and Loss of Supratentorial Myelin

Lorenzo-Ríos et al. 2023
Y. Mah-Som *et al. 2021*

# CEDNIK Syndrome

This disease is caused by a loss-of-function variant in SNAP29 in 22q11.2 region (synaptosomal-associated protein 29), which encodes a SNARE protein, a vesicular membrane fusion protein.

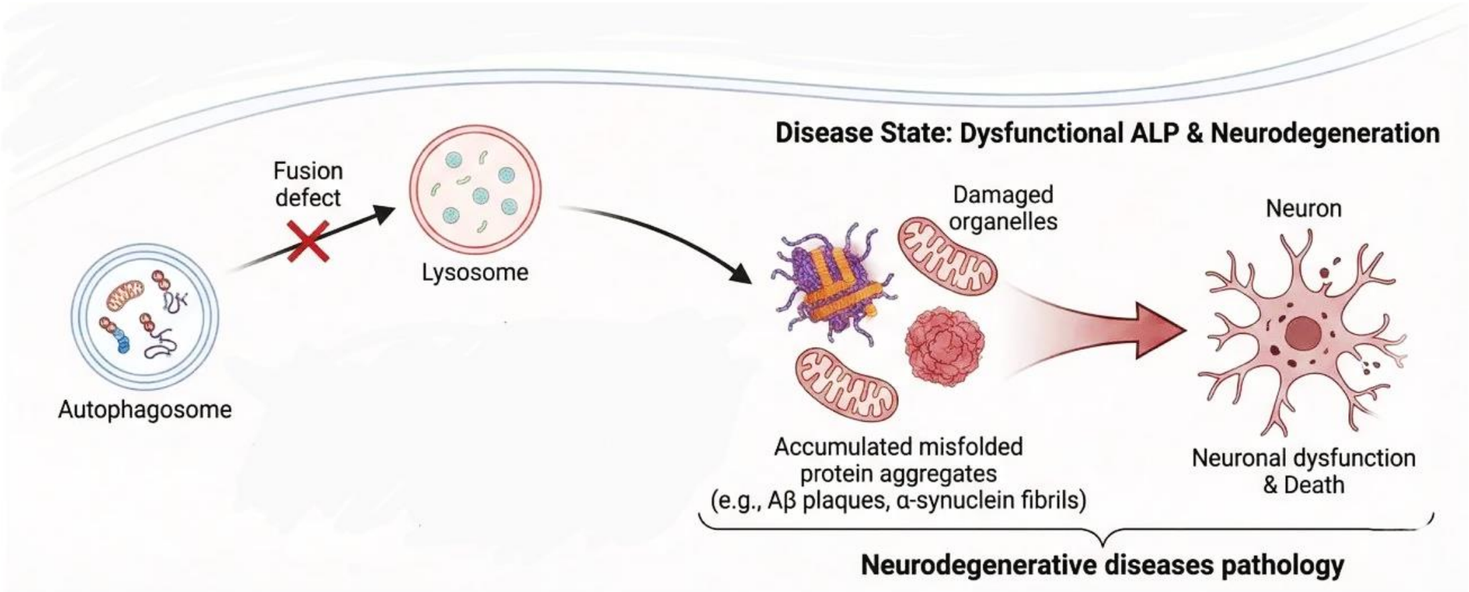



**Figure 3.** The autophagosome-lysosome fusion machinery and the autophagic flux block neuronal protein aggregation

H. Salimi Dafsar et al., 2025 [Adapted]

# CEDNIK Syndrome

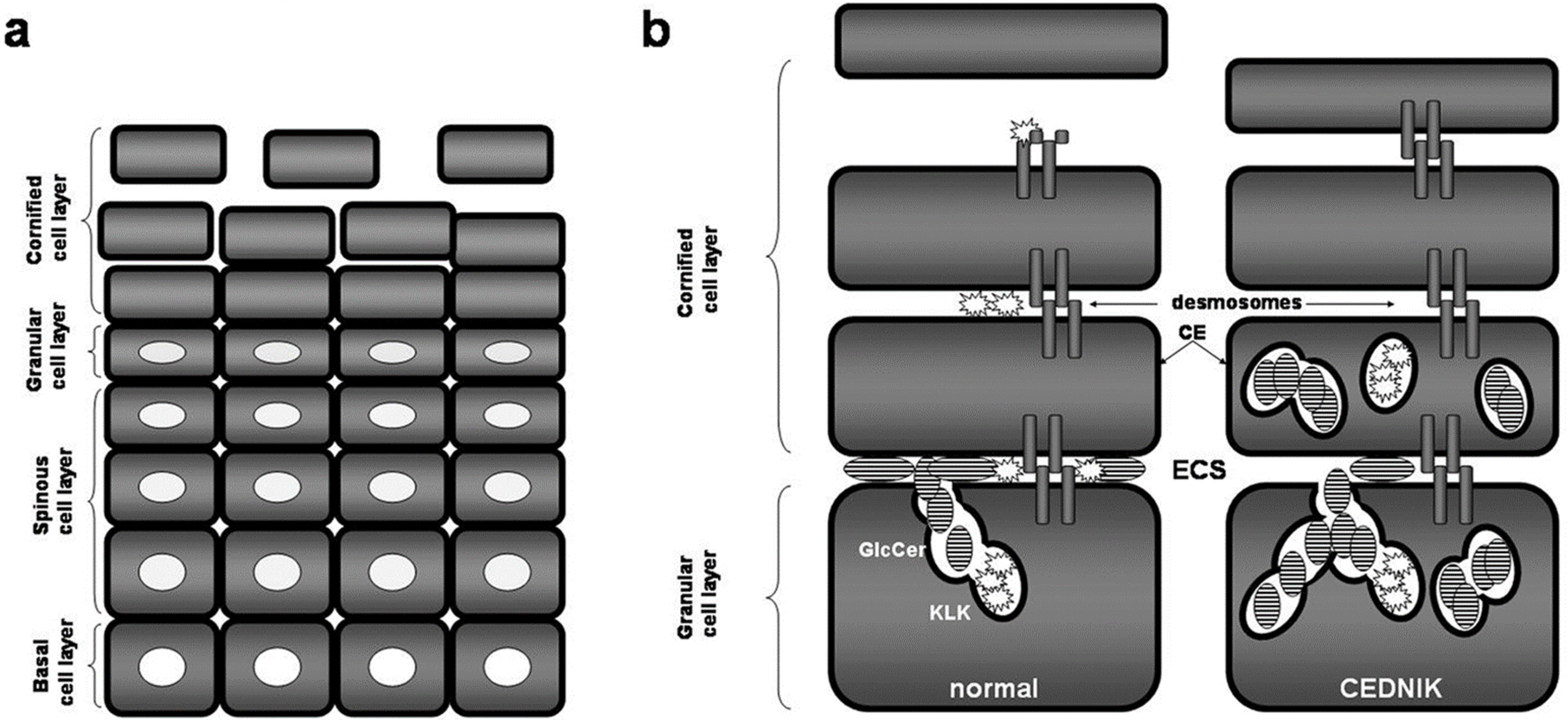



**Figure 4.** CEDNIK molecular mechanisms of epidermal barrier disruption.

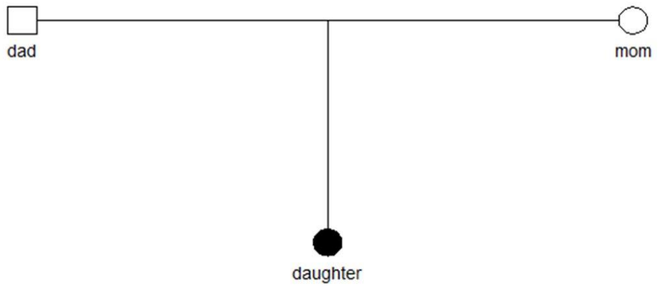
Schiller et al, 2016
https://www.cell.com/cms/10.1086/432556/asset/fa218e55-f090-441e-92c9-984271935749/main.assets/gr6_lrg.jpg

# Symptoms

| Human Phenotype Ontology | Clinical Description |
|---|---|
| HP: 0000252 | Microcephaly |
| HP:0001263 | Global developmental delay |
| HP:0001251 | Ataxia |
| HP:0008064 | Ichthyosis |
| HP:0007447 | Diffuse palmoplantar hyperkeratosis |

# Case

- A 3-year-old girl was referred for severe developmental delay and a progressive skin disorder. She has microcephaly, poor motor development and marked delay of all developmental milestones. Neurological examination shows impaired coordination.
- During the second year of life she developed generalized ichthyosis and pronounced thickening of the palms and soles. Both parents are clinically healthy.




# WES – Raw Data

➢ **The Corpas Project**

Based on the crowdsourced genomic analysis provided by M. Corpas et al. (2015), we used a trio family case study. For the analysis, the data provide was a Whole-Exome Sequencing (WES) along with specific disease-causing variant positions synthetically introduced for educational purposes.

# Sample Filtering WorkFlow

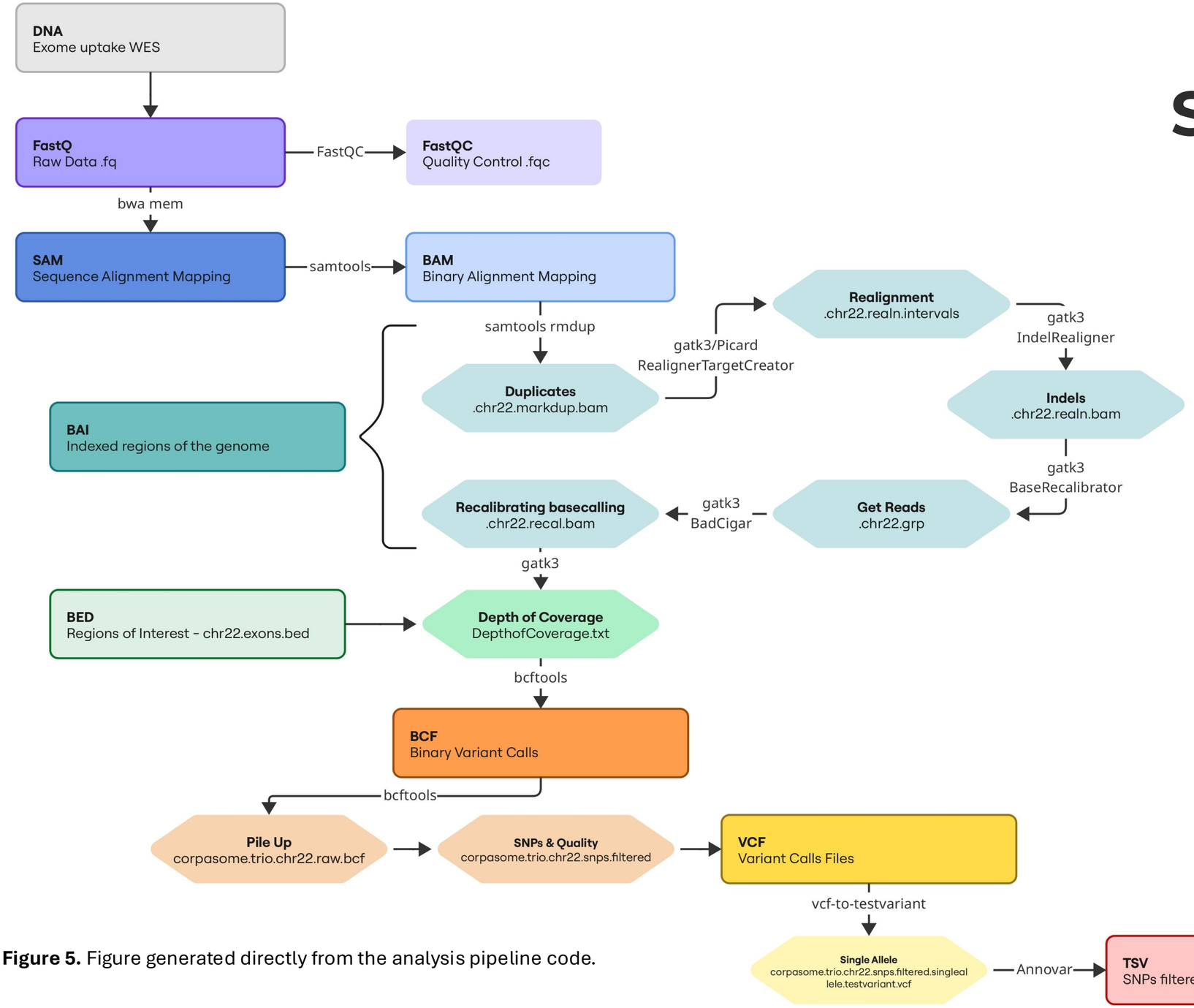



**Figure 5.** Figure generated directly from the analysis pipeline code.

- **hg19**

→ Reference human genome

- **1000g2012apr_eur, esp6500si_ea**

→ 1000 genome project and most common european genes as reference

- **snp138**

→ Known changes at snp's (dbsnp)

- **ljb2_pp2hvar (polyphen), ljb2_sift**

→ Important changes in amino acides

- **clinvar_20140702**

→ Mutations related to diseases

# Quality

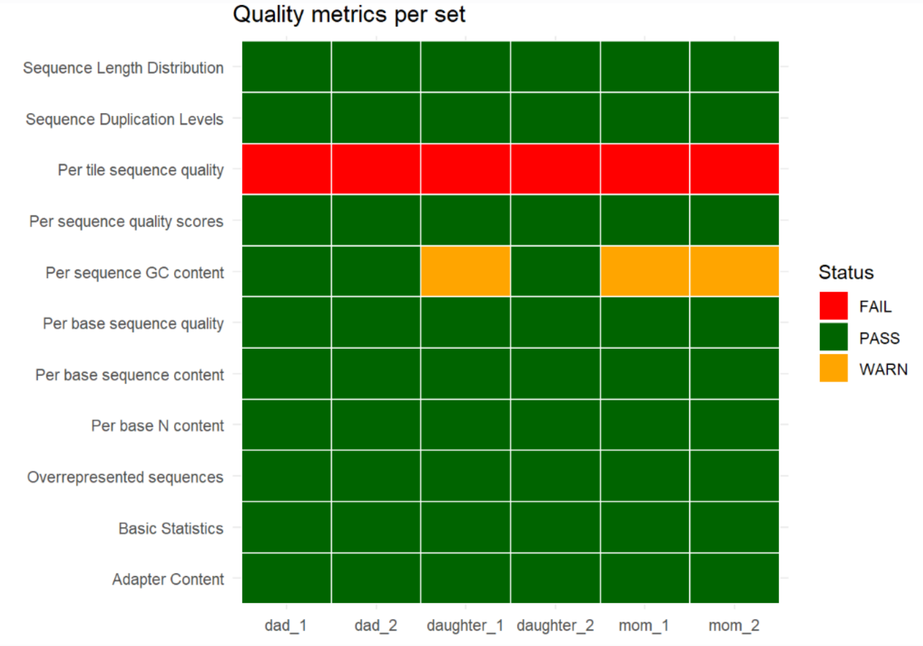
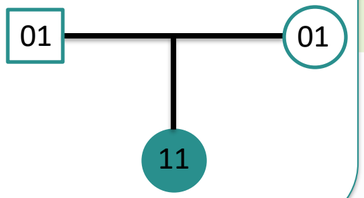



**Figure 6.** FastQC metrics across all samples




**Figure 7.** Read quality metrics.

# Variant filtering

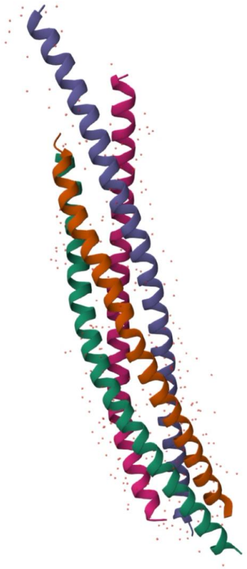
| | 1 Variants | 2 Recessive | 3 Noncommon | 4 Exonic | 5 Deleterious |
|---|---|---|---|---|---|
| *#Variants* | 5983 | 153 | 152 | 37 | 23 |
| *#Genes* | 298 | 61 | 60 | 17 | 11 |
| | corpasome.trio.chr22.snps.filtered.annovar.tsv |   | remove variants that are frequent in population databases | Keep variants in exonic regions only | Filter for deleterious variants |

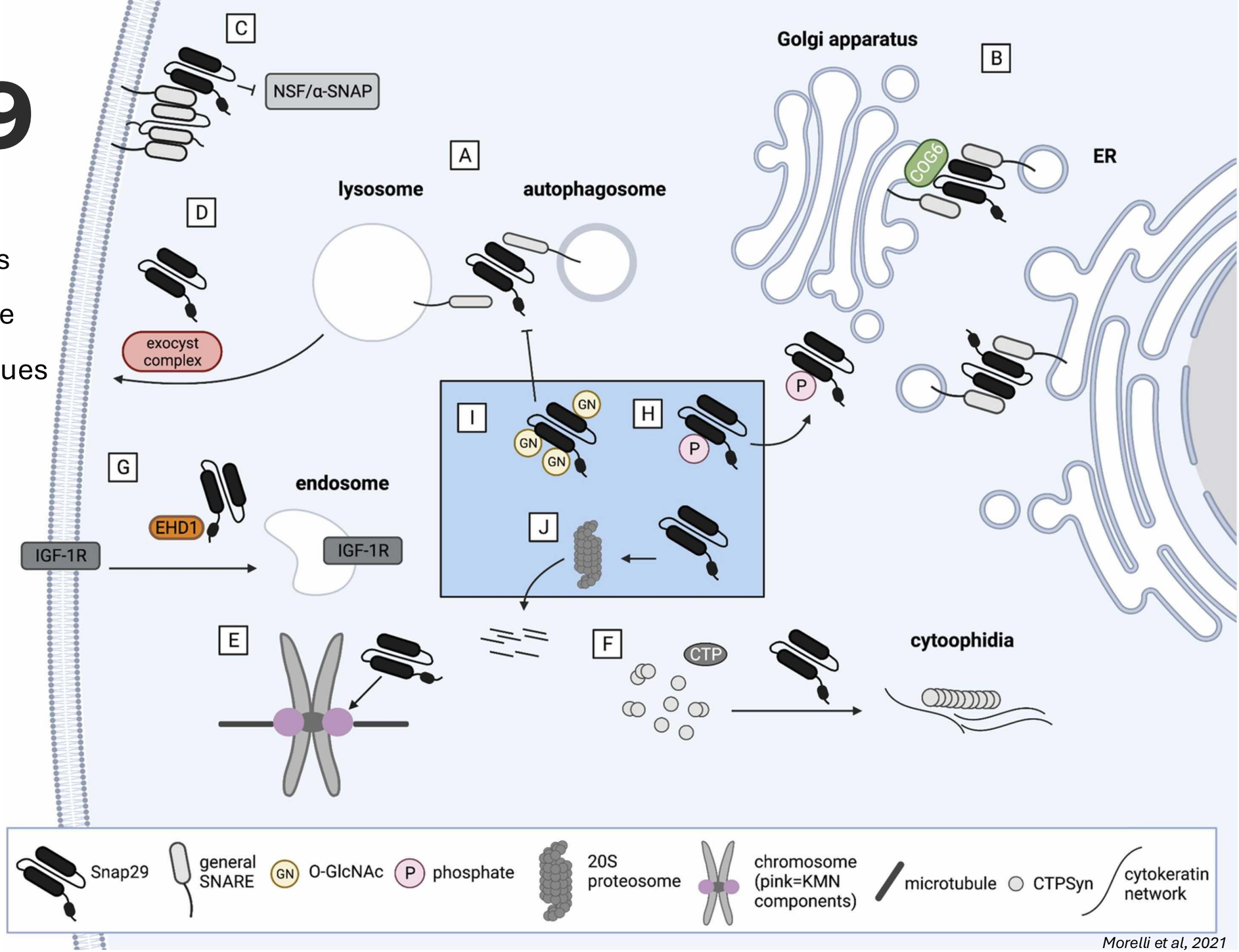
| | |
|---|---|
| No disease association | CCT8L2, ZNF280A |
| Cancer | ATP6V1E1, MED15, RFPL2, XRCC6, NHP2L1, ATP5L2, BRD1, ZBED4 |
| **CEDNIK** | **SNAP29** |

Varsome

# SNAP29

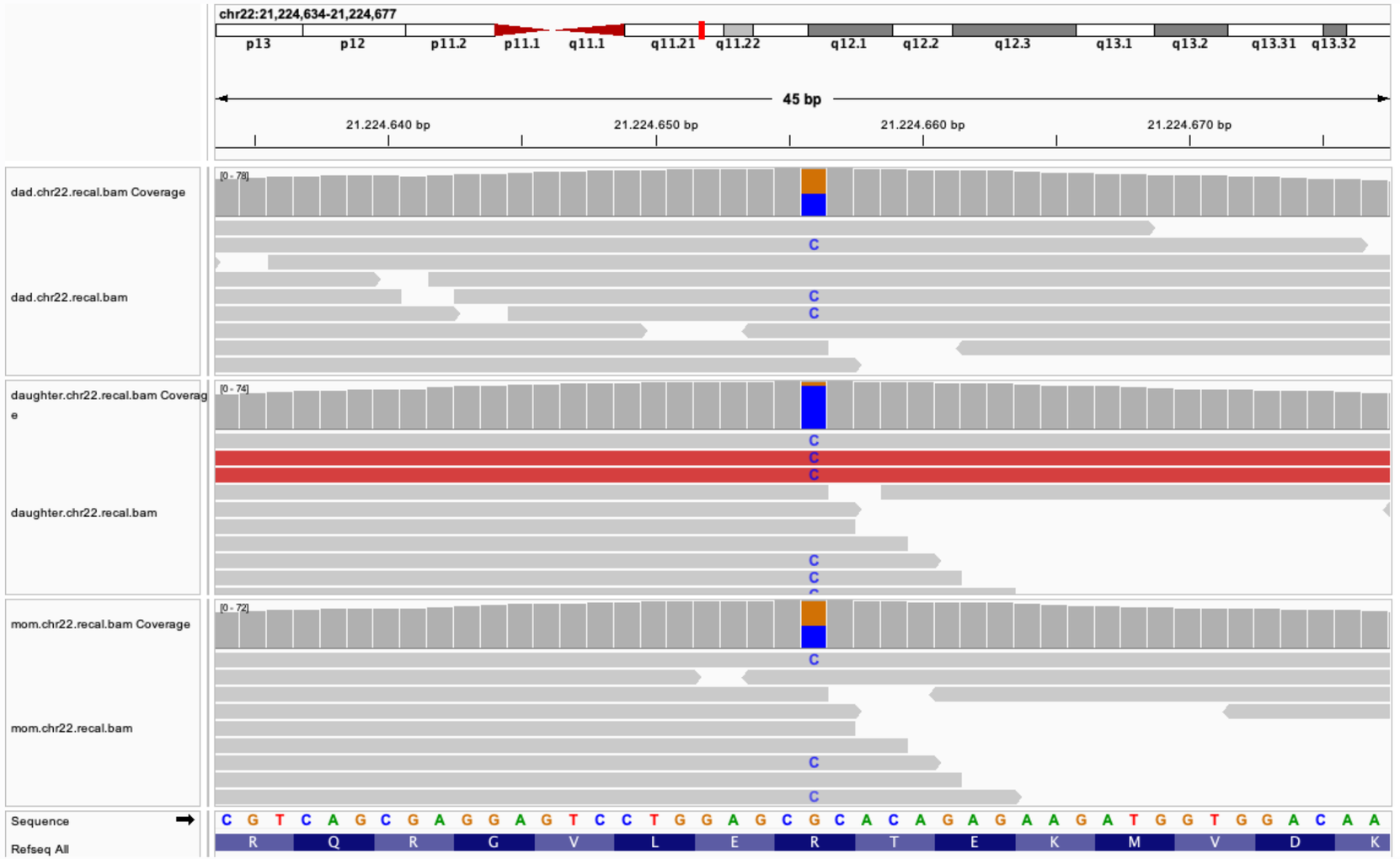
- Family of SNARE proteins
- Conserved trafficking role
- Expressed across all tissues






*Morelli et al, 2021*

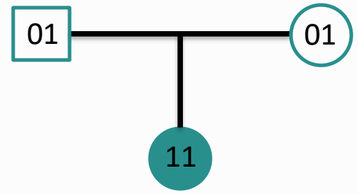
# Our candidate variant

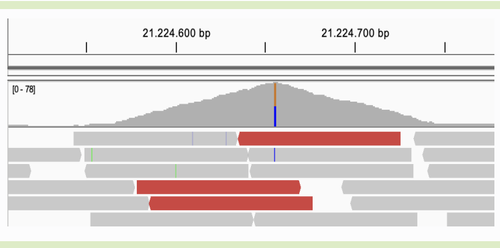



chr22:g.21224656G>C

c.269G>C

p.Arg90Pro



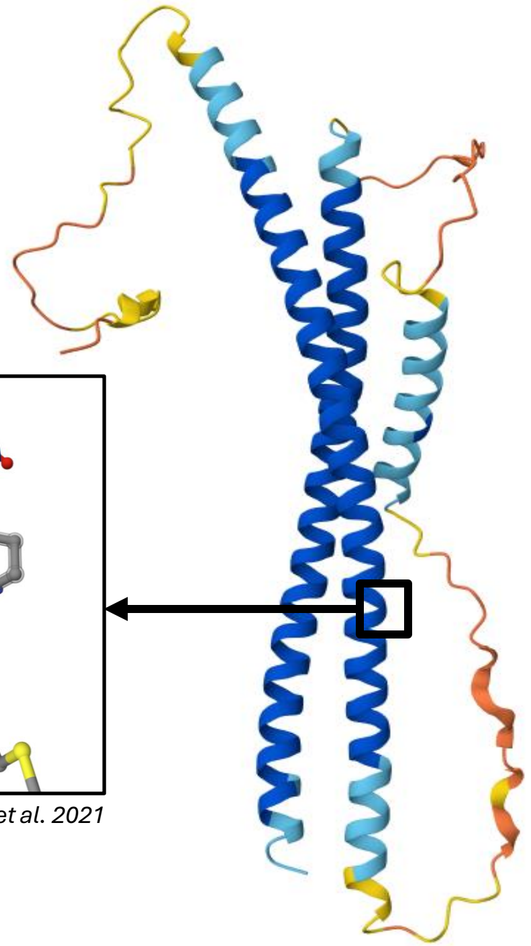

*IGV version 2.19.8*

# Our candidate variant

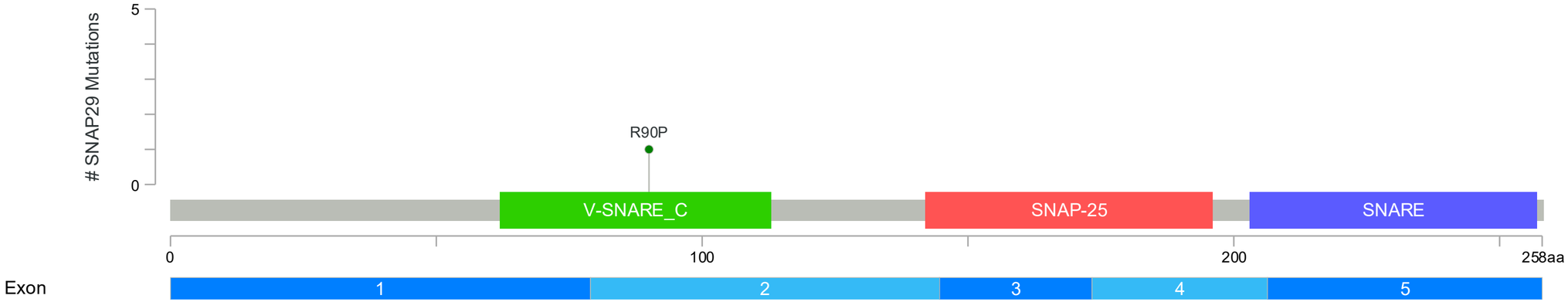
- Variant positioned in alpha helix of V-Snare motif
  - Conserved structure (PhyloP100Way score: 4.128)

- Proline known to break helical structure due to missing NH-group
  - AlphaMissense prediction: “pathogenic moderate“

- Several variants in SNAP29 associated with CEDNIK syndrome



*Jumper et al. 2021*




*cBioPortal v7.1.2*

# Discussion and Improvements

**Database**

Segregation and Inheritance Compatibility

**Reference Databases**

Updated reference genomes. eg hg19 into hg38

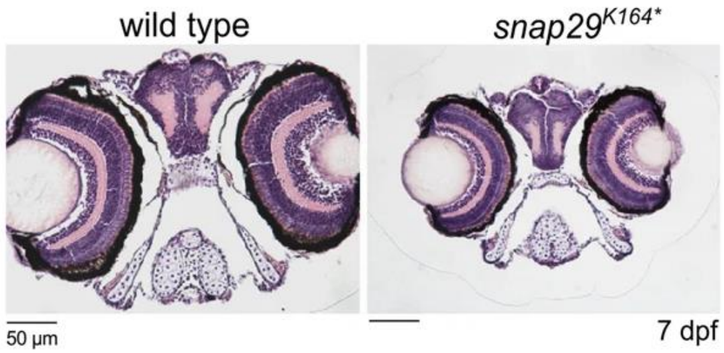
**Homozygosity**

10% of guanine presenting reads at variant position

**Experimental Validation**

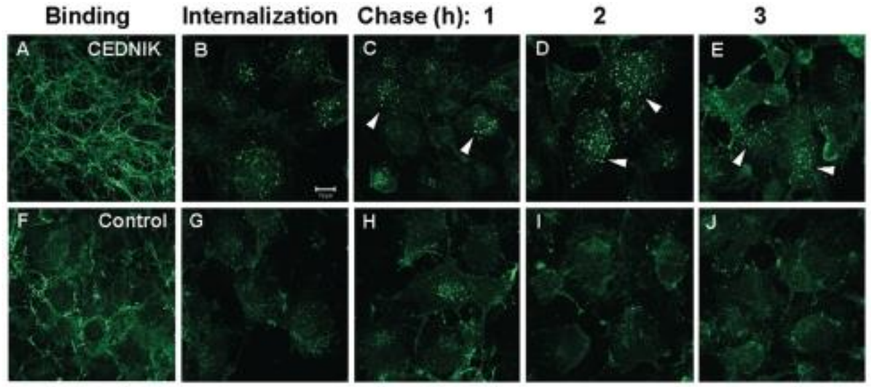
Study the consequences on the protein structure

# Outlook




*Mastrodonato et al, 2019*

- Stained sections of zebrafish larvae
- Reduction in size of the head in *snap29*$^{K164*}$ mutant animals




*Rapaport et al, 2010*

- Analysis of integrin recycling
- Role of integrins in cell migration is essential for epidermal differentiation and brain development
- Accumulation of β-integrin in CEDNIK cells

In patient derived and healthy fibroblasts:

- Analysis of SNAP29 protein binding using immunoprecipitation
- Functional analysis of autophagy (autophagic flux assay)

# References


General Information:

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# References


Tools and Programs:

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2. E. Cerami *et al.*, "The cBio Cancer Genomics Portal: An Open Source Platform for Exploring Multidimensional Cancer Genomics Data," *Cancer Research*, vol. 72, no. 21, pp. 5602–56011, Nov. 2012, doi: 10.1158/0008-5472.CAN-12-0950. cBioPortal v7.1.2

Figures and Graphs:

1. Figure 1: Lorenzo-Ríos D, Guerrero-García A, Colón-Fontánez F. Keratoderma and ichthyosis as valuable features for the diagnosis of CEDNIK syndrome JAAD Case Reports, 2023; 46, 64-66
2. Figure 2: A. Y. Mah-Som et al., "New Cohort of Patients With CEDNIK Syndrome Expands the Phenotypic and Genotypic Spectra," Neurology: Genetics, vol. 7, no. 1, p. e553, Feb. 2021, doi: 10.1212/NXG.0000000000000553
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# Thank You!